

EE2213 Introduction to Artificial Intelligence

Lecture 14

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OVERVIEW OF COURSE CONTENTS

- **Introduction (Shaojing)**

- What is AI
- Applications of AI
- AI agent

- **Search (Shaojing)**

- Uninformed search algorithms: breadth-first, depth-first, uniform-cost(Dijkstra's algorithm)
- Informed search algorithms: greedy best-first, A*
- Applications

- **Optimisation (Shaojing)**

- Linear programming
- Convex problems
- Applications

- **Machine learning (Wang Si)**

- Supervised and unsupervised learning: regression, classification, clustering
- Neural networks and deep learning
- Applications

- **Knowledge representation (Wang Si)**

- Knowledge Representation and Reasoning
- Propositional Logic
- Applications

- **Ethical considerations (Shaojing)**

- Bias in AI
- Privacy concerns
- Societal impact



**WE ARE
HERE**

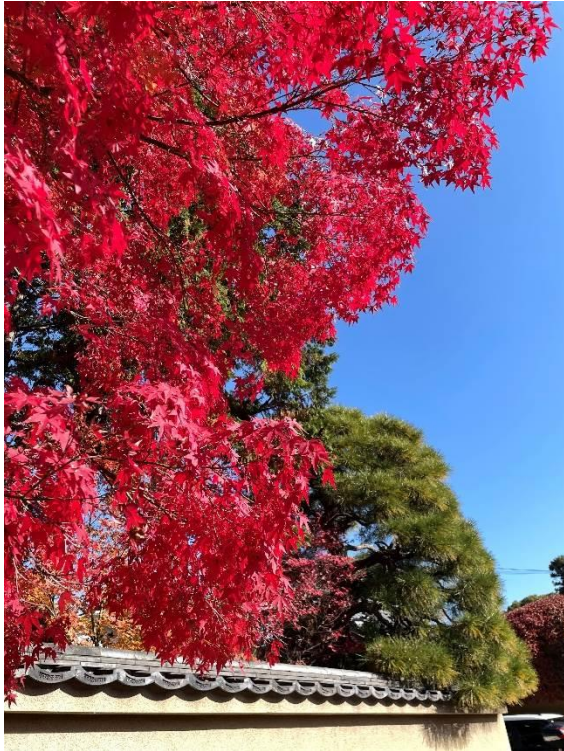
Agenda

- Generative AI & its social impact
- Bias in AI
 - Type of common biases
 - Ways to reduce biases
- Privacy concerns

Can you tell which one(s) was/were created by AI?



Can you tell which one(s) was/were created by AI?



AI content at fingertips



Give me a picture of an autumn scene at Tōfuku-ji Temple with red maple leaves. I want some small human figures in the picture. I want the lighting and illumination to be realistic, like natural light. I also want some blue sky. I want it to look like a real photo taken by a phone.

Gemini

2.0 Flash

Hello, Shaojing

Ask Gemini



Deep Research

Canvas



Can you tell which one(s) was/were created by AI?

Living my dream life in this cozy neighborhood! 🏡 Can you imagine waking up to the sweet scent of oranges every morning? 😊🍊 Who needs a vacation when you have this slice of paradise right at home? 🌴🌟
#DreamHome #OrangeTreeGoals #ParadiseLiving



Extraordinary discovery unveiled: Mysterious small box adorned with enigmatic cat paintings baffles experts



Can you tell which one(s) was/were created by AI?

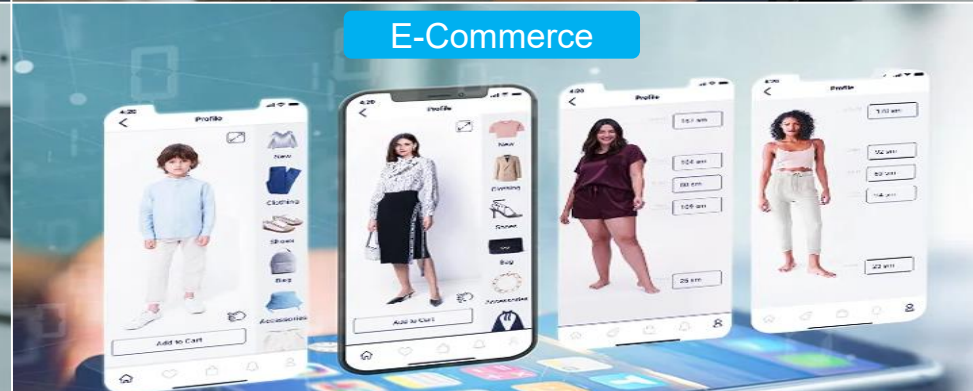
Living my dream life in this cozy neighborhood! 🏡 Can you imagine waking up to the sweet scent of oranges every morning? 😊🍊 Who needs a vacation when you have this slice of paradise right at home? 🌴🌟
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Extraordinary discovery unveiled: Mysterious small box adorned with enigmatic cat paintings baffles experts

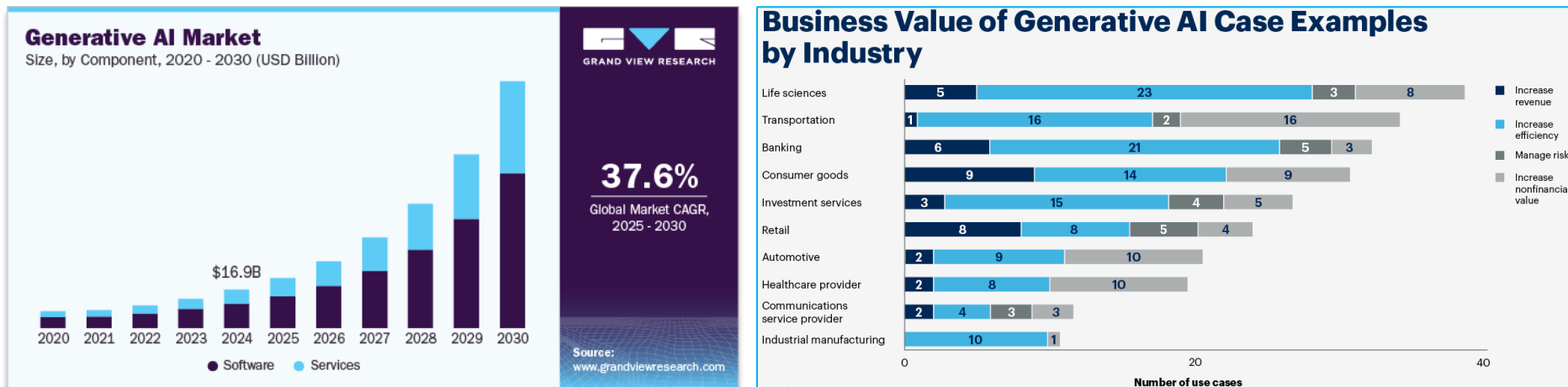


Many facets of AI-generated content



First three images were generated by Google Gemini 2.0 Flash

The growing prevalence of AI-generated content



Source (left): Generative AI Market & Trends, *Grand View Research*, **March, 2025**. URL:

<https://market.us/report/generative-ai-in-content-creation-market/>

Source (right): 2024 generative AI case examples across industries. *Gartner*, *September 2024*. URL:

<https://www.gartner.com/en/articles/generative-ai-use-cases>

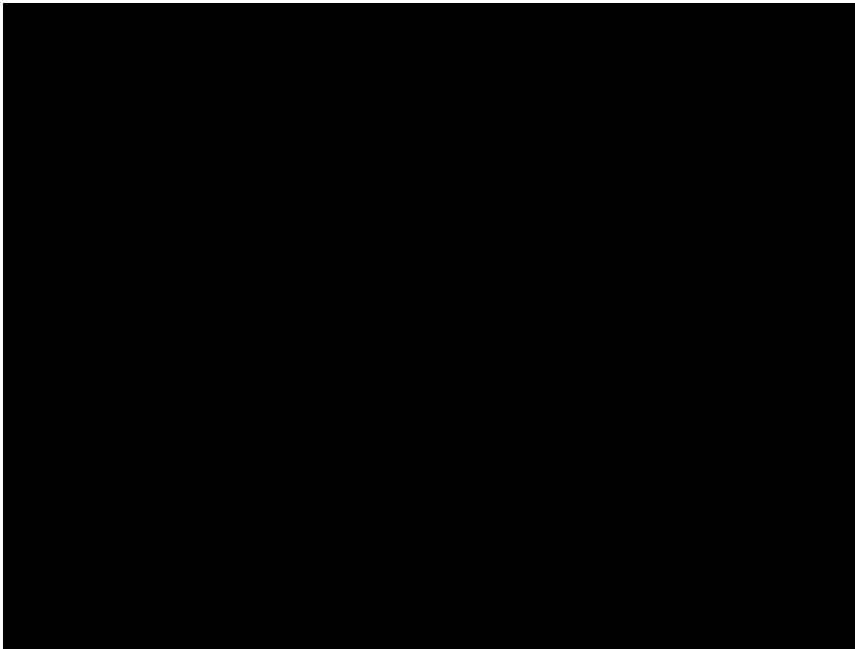
AI: A tool for progress or a misinformation machine?



Success in creating AI could be the biggest event in the history of our civilization. ... **Or the worst.**

— *Stephen* —
*Hawking*²⁰¹⁶

AI: A tool for progress or a misinformation machine?



The 2001 Great Cascadia 9.1 Earthquake & Tsunami - Pacific Coast of US/Canada



Source (left): https://www.tiktok.com/@cycoderx/video/7476805185864486166?_r=1&_t=ZS-8uTRgiMbgNW

Source (right): <https://www.forbes.com/sites/mattnovak/2023/03/27/ai-creates-photo-evidence-of-2001-earthquake-that-never-happened/>

Generative AI: Truth or Bias?

Why Google's AI tool was slammed for showing images of people of colour

Gemini's attempt to correct the marginalisation of ethnic minorities has backfired.



End Wokeness ✓
@EndWokeness



America's Founding Fathers, Vikings, and the Pope according to Google AI:

Certainly! Here is a portrait of a Founding Father of America:



Sure, here is an image of a Viking:



Sure, here is an image of a pope:



Summary: Ethical challenges in Generative AI

- Misinformation and Deepfakes: How generative AI can spread false or harmful content.
- Bias and Fairness: AI models can replicate or amplify societal biases.
- Ownership and Consent: Who owns AI-generated content? Was training data used with permission?

Studying human responses of AI content to combat misinformation



Input a piece of information (headline and image), predict its human responses.

Headline

Corresponding Image

Drop Image Here

- or -

Click to Upload

Image Captioning Model

Blip-2

Explanation LLM

gpt-4o

Password

info@demo

clear

submit

Disclaimer: The predictions provided by this system are generated by artificial intelligence and are based on limited available data and observations. These predictions may not accurately reflect actual human reactions or behaviors. The system makes no promises or guarantees regarding the outcomes of using its product or service. Users are advised to use the predictions for informational purposes only and to exercise their own judgment and discretion when making decisions.

Shen, Zhiqi, Shaojing Fan, Danni Xu, Terence Sim, and Mohan Kankanhalli. "Modeling Human Responses to Multimodal AI Content." *arXiv preprint arXiv:2508.10769* (2025).

Xu, Danni, Shaojing Fan, and Mohan Kankanhalli. "Combating misinformation in the era of generative AI models." *Proceedings of the 31st ACM International Conference on Multimedia*. 2023.

Studying human responses of AI content to combat misinformation

H1: Temporal-Spatial Inconsistency (Account for 34.2%)



Reason 1--Discontinuous motions between current frames and following frames

H3: Gaze Inconsistency (Account for 25.0%)



Reason 6--Abnormal averted gaze from camera

H2: Pixel-Wise Inconsistency (Account for 31.5%)



Reason 2--Resolution: lower on left, higher on right



Reason 3--Light: lighter on left, darker on right



Reason 4--Color: mismatched color in left and right



Reason 5--Artifacts: tampered on left, untouched on others

H4: Body-Face Inconsistency (Account for 7.5%)



Reason 7--Age: older body with younger face



Reason 8--Gender: female body with male face



Juan Hu, Shaojing Fan, and Terence Sim. "Seeing Through Deepfakes: A Human-Inspired Framework for Multi-Face Detection", ICCV, 2025

World-wide initiatives

Partnership on AI (PAI): Responsible Practices for Synthetic Media

Key Practices:

- **Disclosure** – Clearly label AI-generated content
- **Consent** – Use likeness/data only with appropriate permissions
- **Transparency** – Provide metadata or provenance info where possible

Who's Involved:

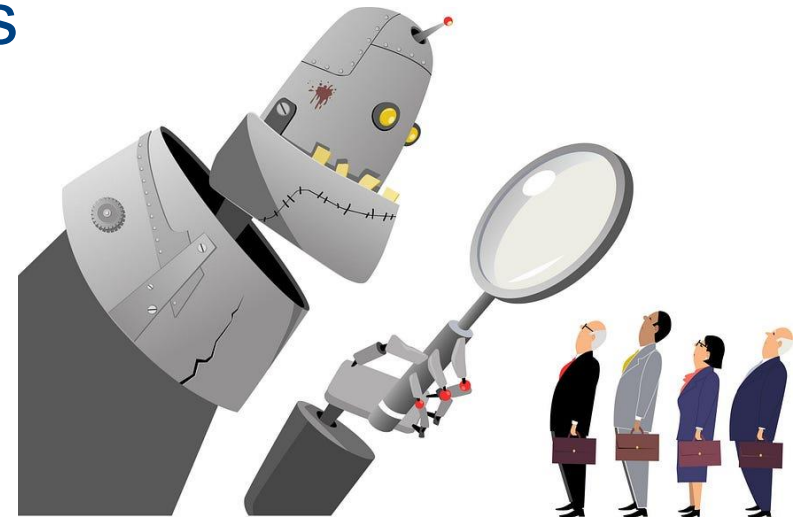
- 50+ global partners, incl. TikTok, OpenAI, BBC, Meta, Microsoft, Adobe
- Civil society orgs (WITNESS, Meedan), researchers, platforms

Website: <https://syntheticmedia.partnershiponai.org/>



Agenda

- Generative AI & its social impact
- Bias in AI
 - Type of common biases
 - Ways to reduce biases
- Privacy concerns



Common types of biases

- **Data bias**

Non-representative/imbalanced training datasets, biased heuristics

- **Model bias**

Simplifying assumptions that don't reflect real-world complexity, or (sometimes) overly complex or overfitted assumptions that reduce generalizability.

- **Deployment bias**

Mismatch between model training conditions and real-world usage, such as applying it in the wrong setting or with different data than it was trained on.

- **Evaluation bias**

Flawed or inconsistent evaluation methods, such as relying on a single train/test split or using metrics that don't capture real-world performance

Example of biased heuristics

Imagine a GPS navigation system trying to find the fastest route from home to school. The system uses a heuristic to estimate travel times.

Suppose this heuristic **ignores traffic in urban areas**, assuming every road allows free-flowing traffic like a highway.

As a result, the algorithm **underestimates the cost of city routes** and thinks they are faster than they actually are.



Example of data bias

I'm building a dog vs. cat classifier, observe my current training data set, do you notice any issues?



Training data



Test results

Example of data bias

AI sentencing cut jail time for low-risk offenders, but study finds racial bias persisted

January 23, 2024 8:00 AM | Keith Brannon kbrannon@tulane.edu

[View PDF](#)



Broken Tools: Vincent Southerland illuminates the problem of bias in the criminal legal system's use of algorithmic tools



Source: <https://news.tulane.edu/pr/ai-sentencing-cut-jail-time-low-risk-offenders-study-finds-racial-bias-persisted>
<https://www.law.nyu.edu/news/ideas/vincent-southerland-algorithmic-tools-bias-criminal-justice-reform>

Takeaway: Reduce data bias in practice

- **Use diverse & balanced data**
Ensure fair representation across all groups.
- **Audit & clean the dataset**
Fix labels, remove noise, and anonymize if needed.
- **Validate across subgroups**
Apply stratified cross-validation and track group-wise metrics.
- **Train with fairness in mind**
Use regularization, reweighting, or fairness-aware loss.

Common sources of bias

- **Data bias**

Non-representative/imbalanced training datasets, biased heuristics

- **Model bias (ML context)**

Simplifying assumptions that don't reflect real-world complexity, or (sometimes) overly complex or overfitted assumptions that reduce generalizability.

- **Deployment bias**

Mismatch between model training conditions and real-world usage, such as applying it in the wrong setting or with different data than it was trained on.

- **Evaluation bias**

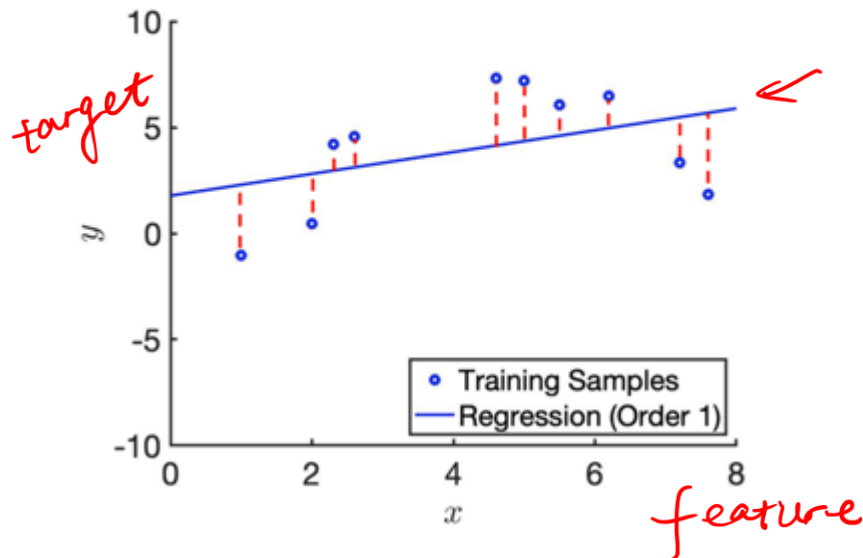
Flawed or inconsistent evaluation methods, such as relying on a single train/test split or using metrics that don't capture real-world performance

Example of model bias

Underfitting occurs when a model is too simplistic to capture the underlying patterns in the data, resulting in poor model performance.

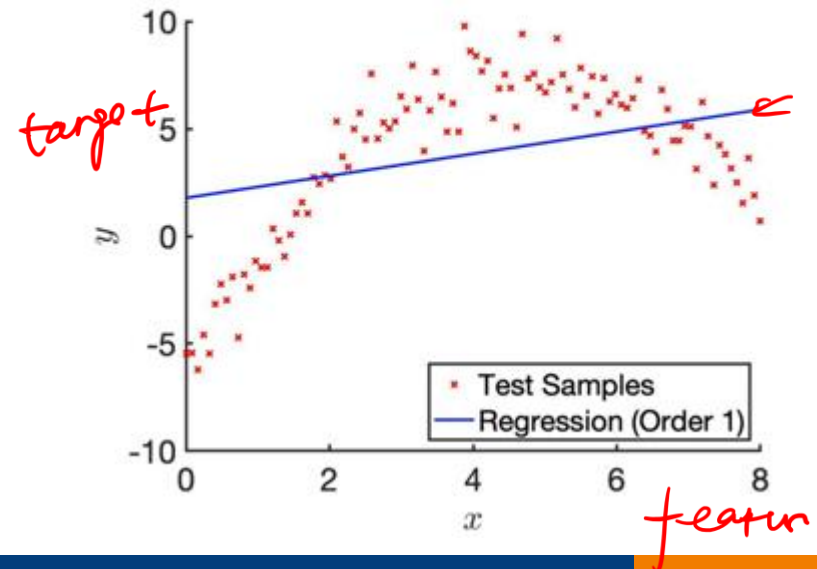
Training

Underfitting Example



Testing

Underfitting Example



Example of model bias

Denied

The Secret Bias Hidden in Mortgage-Approval Algorithms

These two people applied for loans in **Sacramento, Calif.**, in 2019. They both earned **\$70K–\$80K** and sought to borrow **80%** of the property's value.

White applicant **approved**
Asian/Pacific Is. applicant **denied**

These two people applied for loans in **Kansas City, Mo.**, in 2019. They both earned **\$70K–\$75K** and their debt level was **unmanageable**.

White applicant **approved**
Native American applicant **denied**

These two people applied for loans in **Tampa, Fla.**, in 2019. They both earned **\$60K–\$80K** and sought to borrow **95%–100%** of the property's value.

White applicant **approved**
Black applicant **denied**

A loan approval algorithm treats applicants with the same income and credit score as equally creditworthy, **potentially overlooking systemic bias, historical discrimination, and other relevant features.**

Source: <https://themarkup.org/denied/2021/08/25/the-secret-bias-hidden-in-mortgage-approval-algorithms>

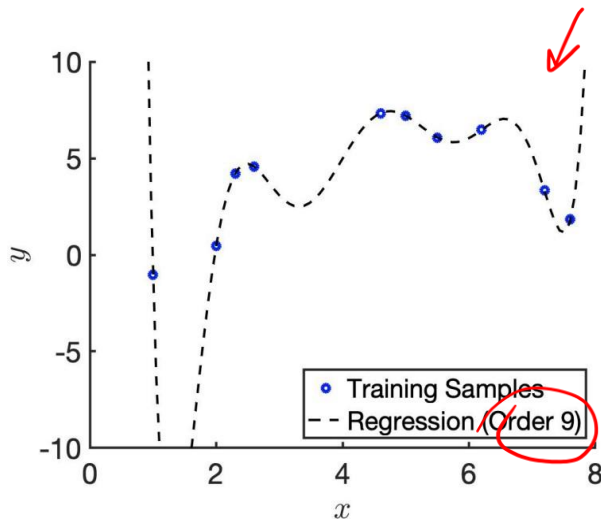
Example of model bias

Overfitting occurs when a model is too complex and learns noise or random fluctuations in the training data, resulting in poor generalization to new data.



Training

Overfitting Example

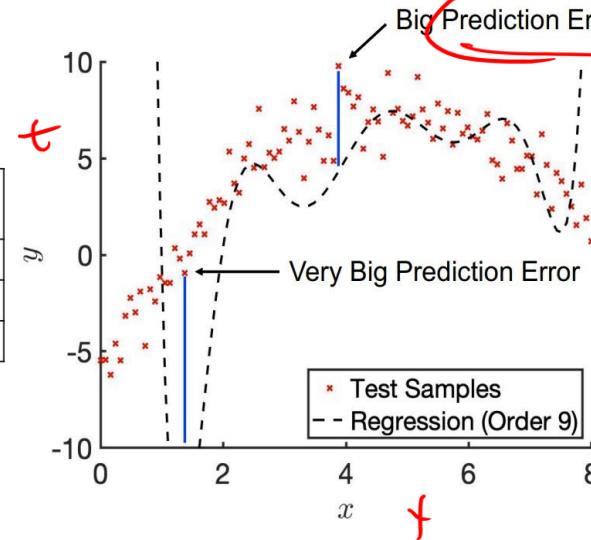


training error = 0

	Training Set Fit	
Order 9	Good	

Testing

Overfitting Example



	Training Set Fit	Test Set Fit
Order 9	Good	Bad

Example of model bias

overfitting



Markets →			Fear & Greed Index →		Latest Market News →
DOW	47,776.16	0.49% ▲			Trump is soon expected to announce
S&P 500	6,877.82	0.04% ▲		Amazon just cut 14,000 jobs, and it's	
NASDAQ	23,712.85	0.32% ▲		CNN charts new course with All Access	

Zillow's home-buying debacle shows how hard it is to use AI to value real estate



By Rachel Metz, CNN Business

7 min read · Published 7:32 AM EST, Tue November 9, 2021



Video Ad Feedback

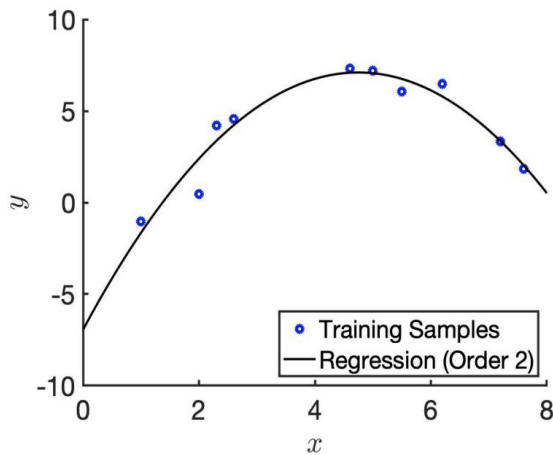
Source: <https://edition.cnn.com/2021/11/09/tech/zillow-ibuying-home-zestimate>

Perfect fitting

The model fits *both training and test* data well.

Training

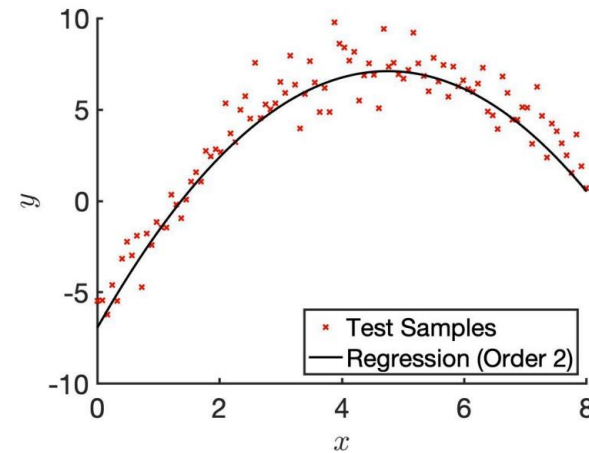
“Just Nice”



	Training Set Fit	Test Set Fit
Order 9	Good	Bad
Order 1	Bad	Bad
Order 2	Good	

Testing

“Just Nice”



	Training Set Fit	Test Set Fit
Order 9	Good	Bad
Order 1	Bad	Bad
Order 2	Good	Good

What causes overfitting (model's perspective)

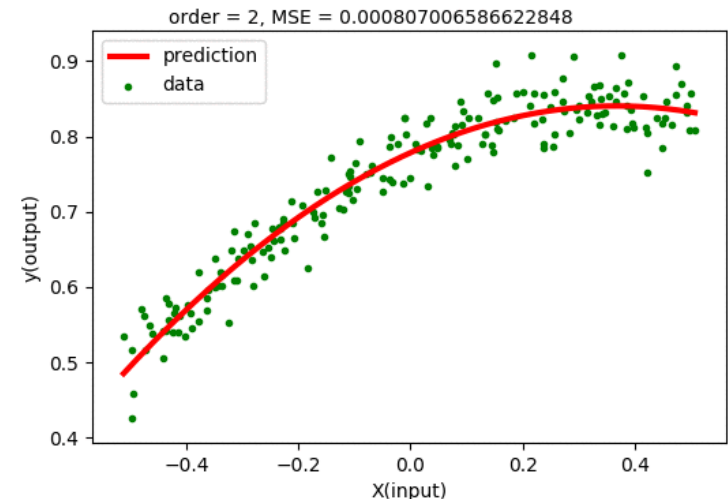
- Possible cause 1: The model is too complex for the given data

An overly complex model tries to capture noise or small fluctuations in the data rather than the underlying pattern.

Solution:

- Use a simpler model (e.g., reduce polynomial order)
- Use regularization (e.g., ridge regression)

Indications of model complexity in our current situation: # of orders; # of parameters; complexity of weight matrix, ...



Source: <https://aiinarabic.com/regression/>

What causes overfitting (data's perspective)

- Possible cause 2: There are too many features and too few training samples.

Application example: Predicting cancer risk based on gene expression

Inputs (Features, X): Gene expression levels of 10,000 genes for each patient.

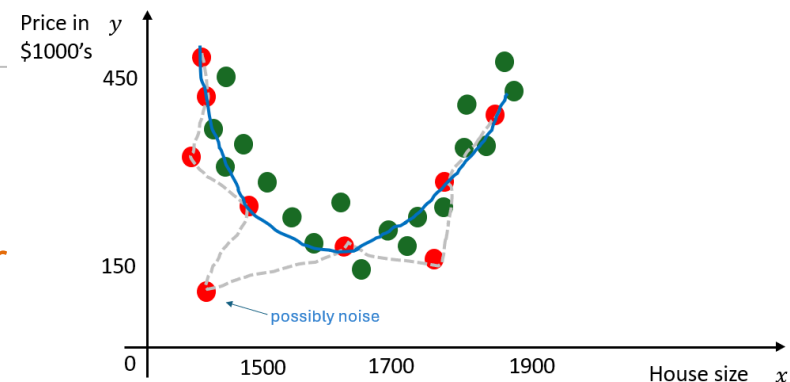
Output (Target, y): A binary variable indicating whether each patient developed cancer (1) or not (0).

$$y = w_0 + w_1x_1 + w_2x_2 + \dots + w_{10000}x_{10000}$$

What do we do if we have an equation with 10,001 parameters and only 500 data points? (under-determined system)

Solution:

- Collect more data
- Reduce the data dimensionality (e.g., feature selection)
- Use regularization

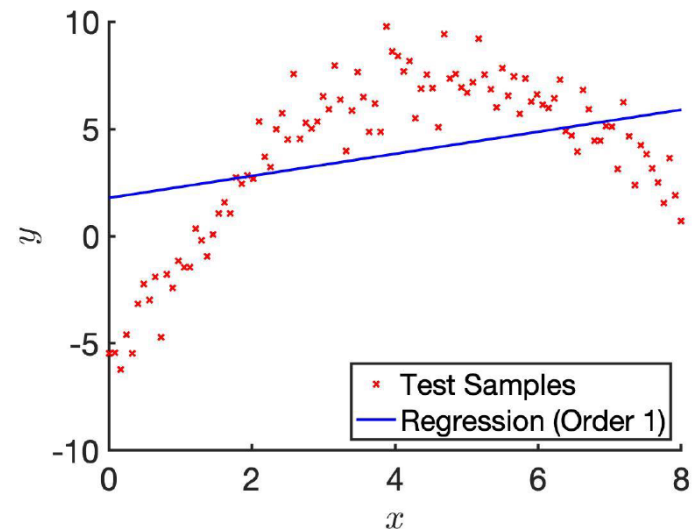


What causes underfitting (model's perspective)

- Possible cause 1: The model is too simple for the given data

Solution:

- Use a more complex model (e.g., increasing the polynomial order, or adopting a more sophisticated algorithm that better captures the data's complexity.)



What causes underfitting (data's perspective)

- Possible cause 2: The features are not informative enough

If the dataset's features lack relevance or meaning, the model may fail to learn useful patterns and instead capture irrelevant information.

Solution:

- Enhance feature engineering, incorporate relevant features, or collect higher-quality data to capture more informative patterns.

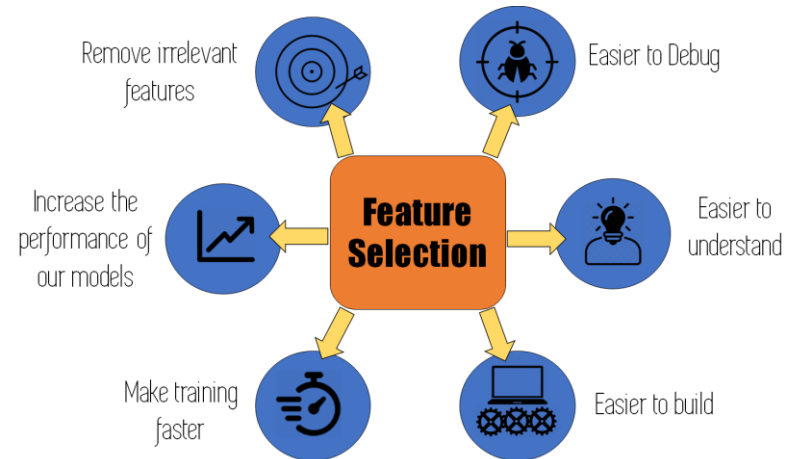


Image credit: <https://medium.com/@dancerworld60/filtering-out-irrelevant-features-a-comprehensive-survey-of-filter-based-techniques-for-feature-9275d86996ab>

Takeaway: Reduce model bias in practice



- **Perform exploratory data analysis**
Detect missing or underrepresented groups and skewed feature distributions early.
- **Apply careful feature selection & engineering**
Include relevant variables, especially those tied to fairness and context.
- **Regularly re-train with updated data**
Refresh the model with diverse, current data to reduce drift and bias over time.

Common sources of bias

- **Data bias**

Non-representative/imbalanced training datasets, biased heuristics

- **Model bias**

Simplifying assumptions that don't reflect real-world complexity.
(Sometimes) overly complex or overfitted assumptions that reduce generalizability.

- **Deployment bias**

Mismatch between model training conditions and real-world usage, such as applying it in the wrong setting or with different data than it was trained on.

- **Evaluation bias**

Flawed or inconsistent evaluation methods, such as relying on a single train/test split or using metrics that don't capture real-world performance

Example of deployment bias: The Challenger Disaster (1986)

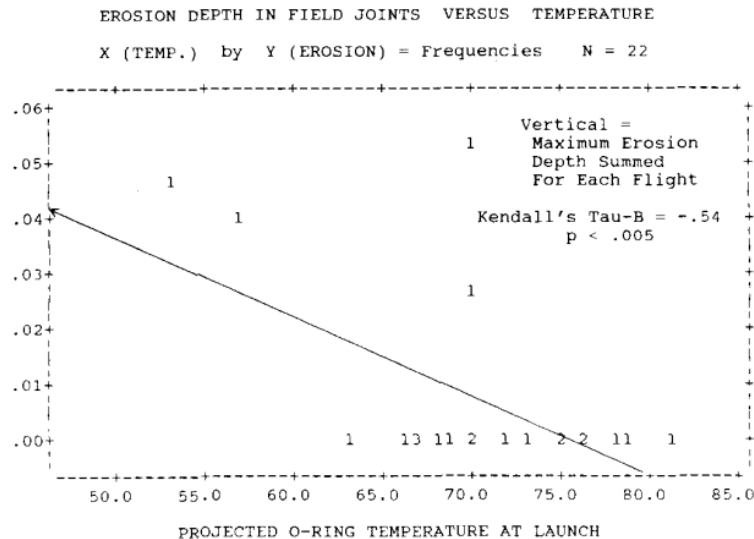
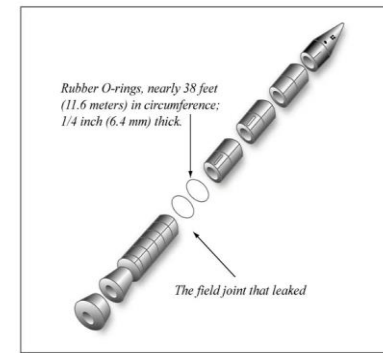
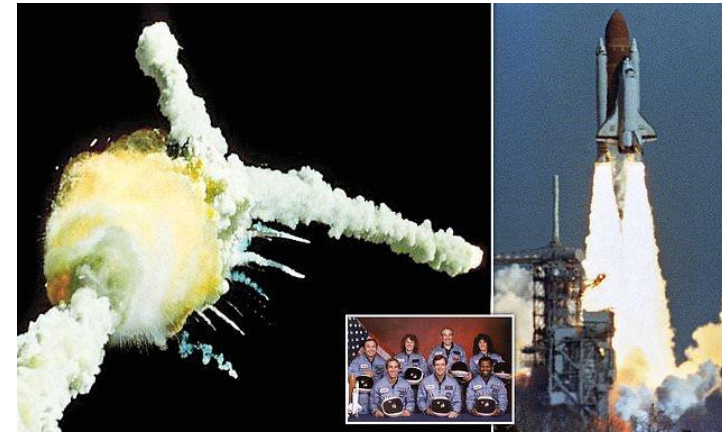


Fig. 3. Linear correlation between O-ring temperature and summed maximum depths (inches) of O-ring erosion for each flight for which complete data were available on January 27, 1986. Frequencies of observations at each bi-variate point in graph are indicated by number.

All the past data came from warmer temperatures (above 53° F or ~12° C).

On launch day, the temperature was unusually low (31° F or ~-1° C).



Courtesy of NASA. Figure by MIT OpenCourseWare

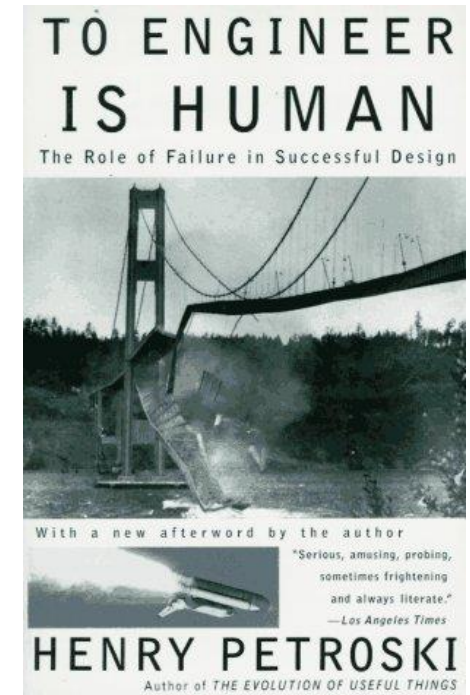
[1] References: Dixon, Robert. "The Challenger Space Shuttle Disaster: A Case Study in the Analysis of Binary Data Using Scatter Diagrams and Logit Regression." *Australian Economic Review* 54.2 (2021): 294-305.

[2] Lighthall, Frederick F. "Launching the space shuttle challenger: disciplinary deficiencies in the analysis of engineering data." *IEEE Transactions on Engineering Management* 38.1 (1991): 63-74.

The Challenger Disaster (1986) : Lessons

Be mindful of your model's **effective range** — predictions made beyond the training data range are **unreliable** and may have serious consequences.

“Examination of testimony and data from the Challenger launch shows that analysis and reasoning were flawed, not due to personal or organizational failings, but to a professional weakness... **a gap in the education of engineers.**”



Lighthall, Frederick F. "Launching the space shuttle challenger: disciplinary deficiencies in the analysis of engineering data." *IEEE Transactions on Engineering Management* 38.1 (1991): 63-74.

Petroski, Henry. "To engineer is human: the role of failure in successful design." *Barnes and novel books* (1985).

Example of deployment bias: Fukushima nuclear accident (2011)



MIT News

ON CAMPUS AND AROUND THE WORLD

SUBSCRIBE

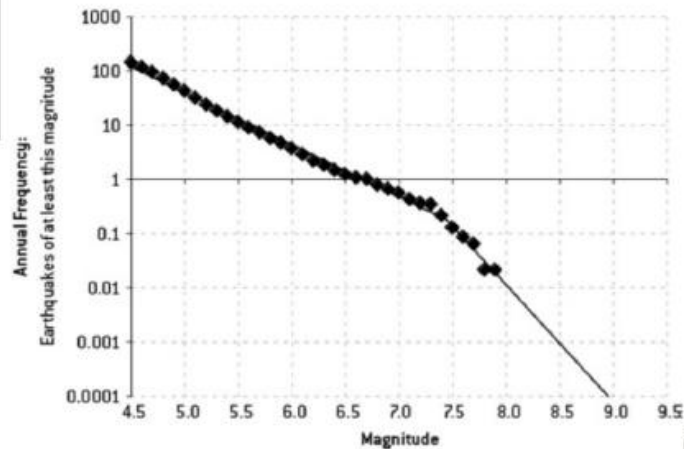
Learning the lessons of Fukushima

Japan's nuclear meltdowns provide valuable lessons for the design of future nuclear powerplants.

David L. Chandler, MIT News Office
August 23, 2011

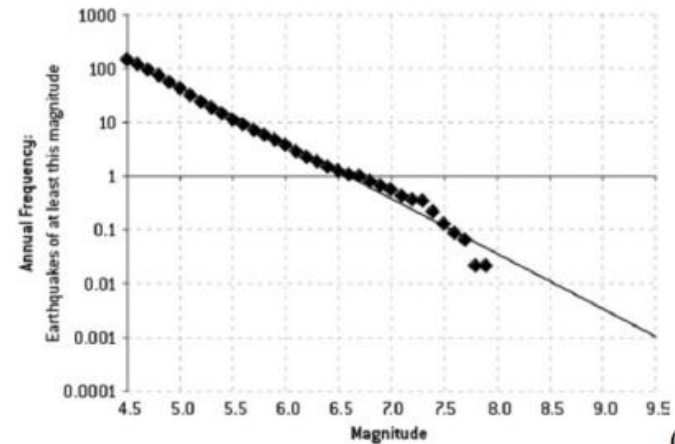


FIGURE 5-7C: TŌHOKU, JAPAN EARTHQUAKE FREQUENCIES
CHARACTERISTIC FIT



(Silver, N, 2012)

FIGURE 5-7B: TŌHOKU, JAPAN EARTHQUAKE FREQUENCIES
GUTENBERG-RICHTER FIT



(Silver, N, 2012)

Reference: N. Silver. The Signal and the Noise: The Art and Science of Prediction. Penguin Books Ltd, 2012.
<https://vyftan.github.io/papers/ee2211book.pdf> P46-47

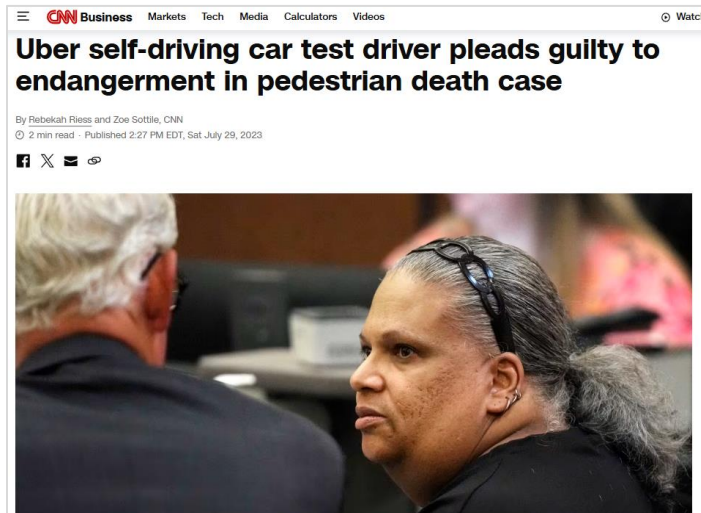
Takeaway: Reduce deployment bias in practice

- Deploy models only in settings that align closely with their training conditions.
- Regularly track model performance and retrain when data or user behavior changes over time.
- Train models on up-to-date, representative datasets that reflect real-world diversity.

Question to ponder



As engineers and scientists, we often focus on building systems that ‘work’—but what happens when those systems are used in ways we didn’t expect? Should engineers be responsible for how their AI models are used in society, even after deployment?



Reference: <https://www.bbc.com/news/business-43459156>
<https://edition.cnn.com/2023/07/29/business/uber-self-driving-car-death-guilty>

Takeaway

Engineering Responsibility in the Real World

"What happens when systems are used in ways we didn't expect?"

- AI models can cause **harm**, even when technically correct
- Engineers play a role in **anticipating risks and misuse**

Ask yourself:

- Who could be harmed by this system?
- What assumptions am I making?
- Have I clearly communicated its limits?

Your responsibility does not end at deployment—ethics must guide design, testing, and real-world impact.

Common sources of bias

- **Data bias**

Non-representative/imbalanced training datasets, biased heuristics

- **Model bias**

Simplifying assumptions that don't reflect real-world complexity.
(Sometimes) overly complex or overfitted assumptions that reduce generalizability.

- **Deployment bias**

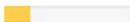
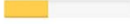
Mismatch between model training conditions and real-world usage, such as applying it in the wrong setting or with different data than it was trained on.

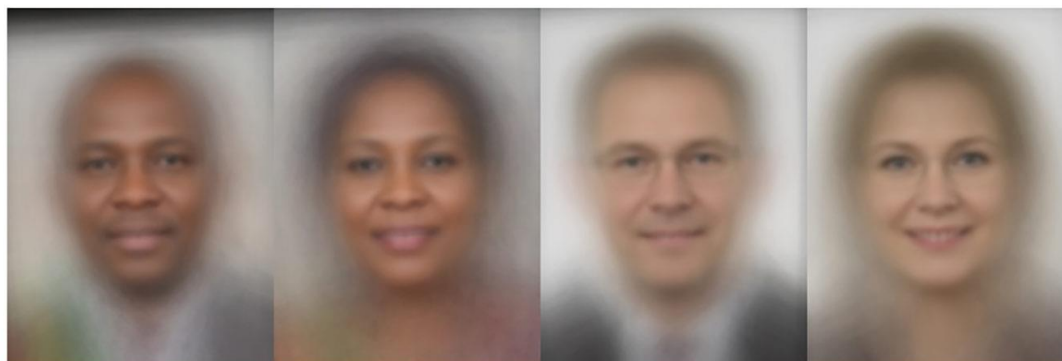
- **Evaluation bias**

Flawed or inconsistent evaluation methods, such as relying on a single train/test split or using metrics that don't capture real-world performance

Example of evaluation bias

Gender Shades study (by Joy Buolamwini and Timnit Gebru at MIT Media Lab)

Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
 Microsoft	94.0% 	79.2% 	100% 	98.3% 	20.8% 
 FACE++	99.3% 	65.5% 	99.2% 	94.0% 	33.8% 
 IBM	88.0% 	65.3% 	99.7% 	92.9% 	34.4% 



References: Buolamwini, Joy, and Timnit Gebru. "Gender shades: Intersectional accuracy disparities in commercial gender classification." *Conference on fairness, accountability and transparency*. PMLR, 2018.

Example of evaluation bias

Suppose we have imbalanced data. Suppose we're building a model to detect diabetes. We've collected data from 1,000 patients, but only 20 are diagnosed with diabetes, while 980 are not.

Suppose our model gets the following result:

	Predicted P	Predicted N
Actual P	10 (TP)	10 (FN)
Actual N	10 (FP)	970 (TN)

In cases of imbalanced data, accuracy alone can be *misleading*. Instead, we can:

- 1) Evaluate using *precision and recall*;
- 2) Apply *cost-sensitive metrics*

Accuracy: $(10+970)/1000 = 0.98$

However, despite the high accuracy, the model correctly identified only **half of the actual diabetes patients**.

The model is **good at predicting negatives** but **struggles with positives**.

Recap: Precision and Recall

Confusion Matrix for Binary Classification

	$\hat{P}$ (predicted)	$\hat{N}$ (predicted)	
P (actual)	TP	FN	Recall $TP/(TP+FN)$
N (actual)	FP	TN	
	Precision $TP/(TP+FP)$		Accuracy $(TP+TN)/(TP+TN+FP+FN)$

Recap: Precision and Recall (cont.)

$$\text{Precision} = \frac{TP}{TP + FP}$$

Correctly predicted positives

Total predicted positives

$$\text{Recall} = \frac{TP}{TP + FN}$$

Correctly predicted positives

Total actual positives

Precision

Prediction

	+	-
Actual +	TP (True Positive)	FN (False Negative)
Actual -	FP (False Positive)	TN (True Negative)

Recall

Prediction

	+	-
Actual +	TP (True Positive)	FN (False Negative)
Actual -	FP (False Positive)	TN (True Negative)

Evaluation Metrics: Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

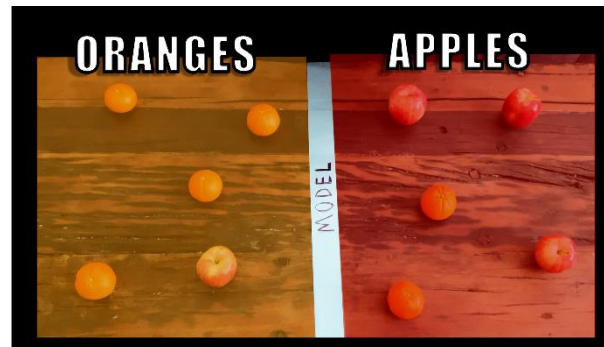


Precision (How accurate the positives are): It measures the quality of model predictions for one particular class. If in a spam detector, precision measures how many of the emails flagged as spam are actually spam. High precision means fewer false alarms.

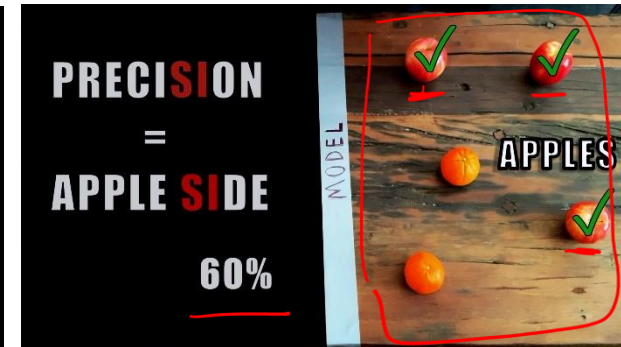
Precision focuses on the reliability of positive predictions, important when the cost of false positives is high.



data: 4 apples (positive class),
6 oranges (negative class)



model prediction results



precision (correctly predicted
positive/total predicted positive)

$$\frac{3}{5} = 60\%$$

Evaluation Metrics: Recall

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

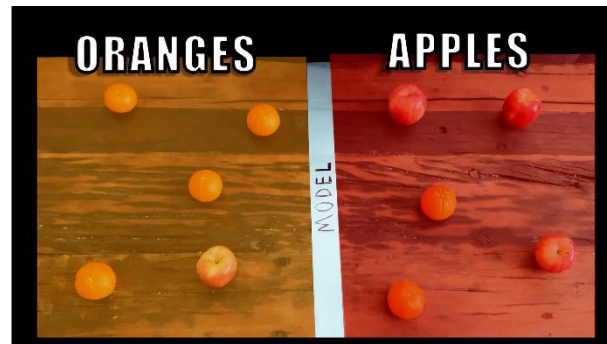


Recall (How many actual positives are detected): If in a spam detector, recall measures how many of the actual spam emails are correctly caught. *High recall means fewer missed cases.*

Recall measures the model's **ability to capture actual positives**, crucial when the **cost of false negatives is high** (e.g., cancer screening, terrorist detection).



data: 4 apples (positive class),
6 oranges (negative class)



model prediction results



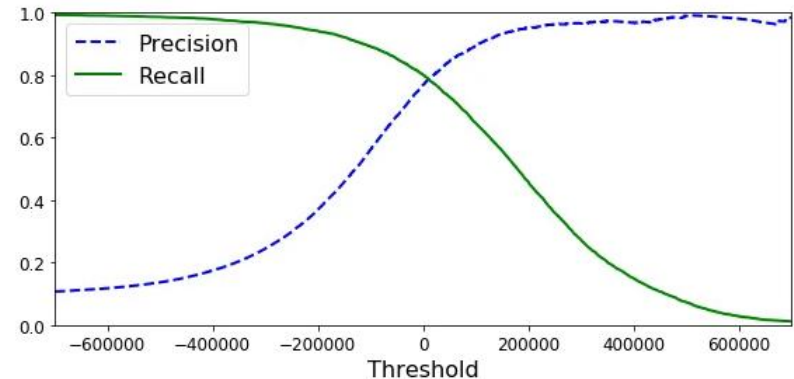
recall (correctly predicted
positive/total **actual** positive)

$$\frac{3}{4} = 75\%$$

Evaluation Metrics: Precision and Recall

In practice, there's often a *trade-off*:

- **Prioritizing precision** reduces **false positives** (e.g., ensuring only actual spam emails are filtered out).
- **Prioritizing recall** reduces **false negatives** (e.g., catching as much spam as possible, even at the risk of misclassifying some legitimate emails as spam).



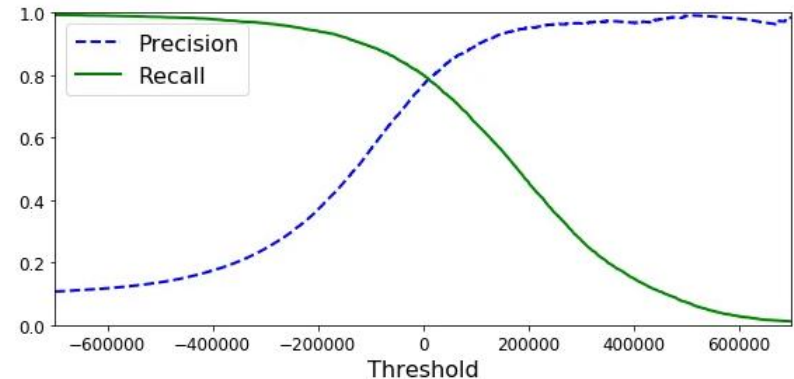
Class practice: Consider these real-life application examples: should we prioritize precision or recall?

- Spam detection
- Cancer diagnosis
- Seismic risk assessment for buildings
- Defect detection in manufacturing
- Autonomous vehicle pedestrian detection

Evaluation Metrics: Precision and Recall

In practice, there's often a *trade-off*:

- **Prioritizing precision** reduces **false positives** (e.g., ensuring only actual spam emails are filtered out).
- **Prioritizing recall** reduces **false negatives** (e.g., catching as much spam as possible, even at the risk of misclassifying some legitimate emails as spam).



Class practice: Consider these real-life application examples: should we prioritize precision or recall?

- Spam detection (prioritize precision)
- Cancer diagnosis (prioritize recall)
- Seismic risk assessment for buildings (prioritize recall)
- Defect detection in manufacturing (prioritize recall)
- Autonomous vehicle pedestrian detection (prioritize recall)

Reduce evaluation bias using precision and recall

Suppose we have imbalanced data. Suppose we're building a model to detect diabetes. We've collected data from 1,000 patients, but only 20 are diagnosed with diabetes, while 980 are not. Suppose our model gets the following result:

	Predicted P	Predicted N
Actual P	10 (TP)	10 (FN)
Actual N	10 (FP)	970 (TN)

Accuracy: $(10+970)/1000 = 0.98$

F1 score (ranges from 0 to 1): a metric that measures a model's accuracy in terms of both precision and recall.

0: the model completely fails to identify the positive class.

1: the model perfectly predicts all positives.

Evaluate the model's performance using precision and recall:

Precision = $10/10 = 0.5$

Recall = $10/10 = 0.5$

$$F1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = 0.5$$

Cost sensitive evaluation metrics

Cost Matrix for Binary Classification

	$\hat{P}$ (predicted)	$\hat{N}$ (predicted)
P (actual)	$C_{p,p} * TP$	$C_{p,n} * FN$
N (actual)	$C_{n,p} * FP$	$C_{n,n} * TN$

Total cost:

$$\begin{aligned} &C_{p,p} * TP + \\ &C_{p,n} * FN + \\ &C_{n,p} * FP + \\ &C_{n,n} * TN \end{aligned}$$

Main idea of cost matrix:

It assigns different penalties for different types of outcomes.

Higher penalties for more severe errors.

Usually, $C_{p,p}$ and $C_{n,n}$ are set to 0; $C_{n,p}$ and $C_{p,n}$ may and may not be equal.

Cost sensitive evaluation metrics example

Evaluate the model's performance using cost matrix (conceptual)

	Predicted P	Predicted N
Actual P	10 (TP)	10 (FN)
Actual N	10 (FP)	970 (TN)

Cost Matrix for Binary Classification

	$\hat{P}$ (predicted)	$\hat{N}$ (predicted)
P (actual)	$C_{p,p} * TP$	$C_{p,n} * FN$
N (actual)	$C_{n,p} * FP$	$C_{n,n} * TN$

Total cost:
 $C_{p,p} * TP +$
 $C_{p,n} * FN +$
 $C_{n,p} * FP +$
 $C_{n,n} * TN$

Original accuracy: $(10+970)/1000 = 0.98$

But if we care more about false negatives (FN), we can assign a higher penalty, e.g., $C_{p,n} = 100$.

The cost-sensitive accuracy drops to $(10+970)/(10 + 10*100+10+970) = 0.49$

Class practice: $C_{n,p} > C_{p,n}$ or $C_{n,p} < C_{p,n}$

Cost Matrix for Binary Classification

	$\hat{P}$ (predicted)	$\hat{N}$ (predicted)
P (actual)	$C_{p,p} * TP$	$C_{p,n} * FN$
N (actual)	$C_{n,p} * FP$	$C_{n,n} * TN$

Total cost:

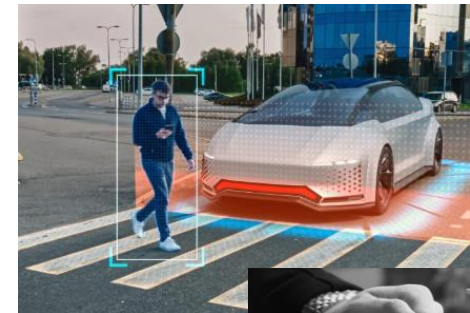
$$C_{p,p} * TP + C_{p,n} * FN + C_{n,p} * FP + C_{n,n} * TN$$

Pedestrian detection in self-driving car systems

Credit card fraud detection

Automated candidate screening systems for universities/companies

Security screening system at customs/airports



Another example of evaluation bias

Suppose we are building a dog/cat classifier. The training and test sets are as shown below. What potential problems could affect model evaluation in this scenario?



Train set



Test set

This could potentially cause **evaluation bias due to an unrepresentative test set**
Relying on a single train/test split can lead to evaluation bias

Reduce evaluation bias using K-fold cross-validation

K-Fold Cross-Validation

Problem: One train–test split may give misleading results, as the chosen test set may not represent the whole dataset

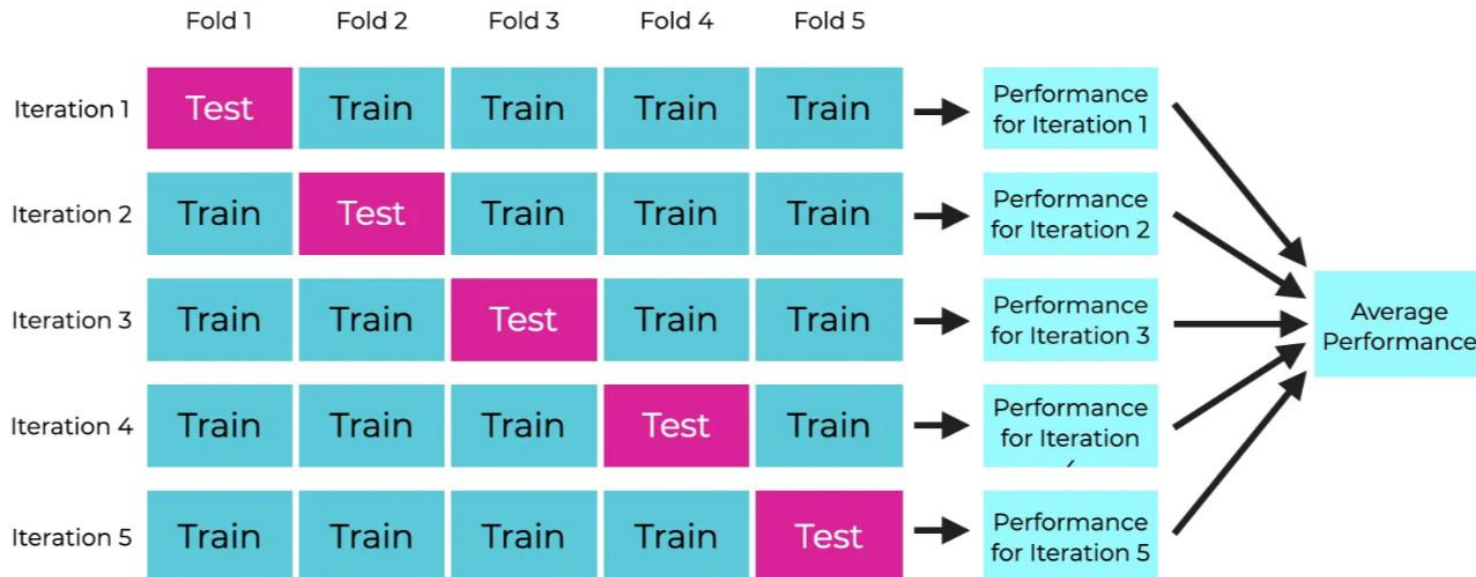
Solution: K-Fold Cross-Validation

Split the dataset into k equal folds.

Train on $(k-1)$ folds, test on the remaining fold.

Repeat k times, each fold serving once as the test set.

Average the results for a more reliable performance estimate.



Takeaway: Engineering practices to reduce AI bias



1. Examine your data thoroughly

Use exploratory data analysis to uncover imbalances, missing groups, or skewed distributions that may introduce bias early on.

2. Design models with real-world complexity in mind

Avoid overly simplistic assumptions—include relevant features like context, fairness factors, and user diversity.

3. Perform robust validation and lifecycle monitoring

Conduct stratified cross-validation across diverse subpopulations and deployment conditions. Implement continuous monitoring to detect distribution shifts and feedback loops that may amplify bias over time.

4. Know where your model works (and doesn't)

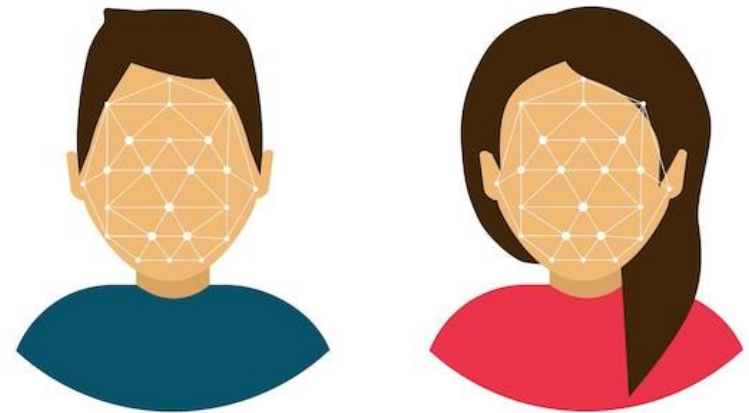
Be clear about the model's effective range; avoid deploying it beyond its design scope.

5. Evaluate your model using k-fold cross validation, or cost sensitive metrics whenever necessary.

Your responsibility does not end at deployment—ethics must guide design, testing, and real-world impact.

Agenda

- Generative AI & its social impact
- Bias in AI
 - Type of common biases
 - Ways to reduce bias
- **Privacy concerns**



Data privacy issues in LLMs

Wiz Research Uncovers Exposed DeepSeek Database Leaking Sensitive Information, Including Chat History

A publicly accessible database belonging to DeepSeek allowed full control over database operations, including the ability to access internal data. The exposure includes over a million lines of log streams with highly sensitive information.



Gal Nagli

January 29, 2025

3 minute read



Forbes

DeepSeek Data Leak Exposes 1 Million Sensitive Records

By [Lars Daniel](#), Contributor. © Lars Daniel covers digital evidence and forensi...

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Published Feb 01, 2025, 08:27pm EST, Updated Feb 02, 2025, 11:15am EST

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Source: <https://www.forbes.com/sites/larsdaniel/2025/02/01/deepseek-data-leak-exposes--1000000-sensitive-records/>
<https://www.wiz.io/blog/wiz-research-uncovers-exposed-deepseek-database-leak>

Efforts to address data privacy issues in LLMs

- **Anonymized training data**
Remove personal identifiers before model training.
- **Differential privacy**
Add noise to protect individual data in learning/output.
- **Access controls & filters**
Restrict sensitive content during model use.
- **On-device / federated learning**
Keep data local; train without direct access.

Data protection against deepfakes



Source: <https://www.youtube.com/shorts/0dTmRCHdXT8>

Privacy issues on social media

Forbes

Clearview AI—Controversial Facial Recognition Firm—Fined \$33 Million For ‘Illegal Database’

By **Robert Hart**, Former Staff. Robert Hart is a former London-based Forbes senior reporter.

Published Sep 03, 2024, 07:54am EDT

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TOPLINE

Controversial U.S. facial recognition company Clearview AI, reportedly embraced U.S. government and law enforcement agencies, has been **fined** more than \$30 million by the Netherlands’ data protection watchdog on Tuesday for building “an illegal database” containing billions of faces taken from social media and the internet.



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Source: <https://www.forbes.com/sites/roberthart/2024/09/03/clearview-ai-controversial-facial-recognition-firm-fined-33-million-for-illegal-database/>

Privacy protection on social media

	Original image		Perturbed image	
				
	Human judgment	Machine judgment	Human judgment	Machine judgment
People present?	✓	✓	✓	✗
Text present?	✓	✓	✓	✗
Contain focused object?	✓	✓	✓	✓
Aesthetic?	✓	✓	✓	✓
Pleasant?	✓	✓	✓	✓

Shen, Zhiqi, Shaojing Fan, Yongkang Wong, Tian-Tsong Ng, and Mohan Kankanhalli. "Human-imperceptible privacy protection against machines." In *Proceedings of the 27th ACM international conference on multimedia*, pp. 1119-1128. 2019.

Summary: Societal impact and more



Bias mitigation via diverse data and fairness-aware models



Minimize risks and maximize AI benefits



Enhancing content moderation for information integrity



Privacy-preserving methods in data generation and transmission



AI, created by humans, serves us better when guided by human insights.

Further readings

AI Bias: 14 Real AI Bias Examples & Mitigation Guide

<https://www.crescendo.ai/blog/ai-bias-examples-mitigation-guide>

AI or human bias?

<https://medium.com/@turalt/ai-isnt-biased-we-are-b74ec94d1698>

AI dilemma:

https://www.youtube.com/watch?v=cB0_-qKbaI4

AI ethics in healthcare research paper:

Hanna, Matthew G., et al. "Ethical and bias considerations in artificial intelligence/machine learning." *Modern Pathology* 38.3 (2025): 100686.

